Executive Overview

From historical demand to inventory decisions

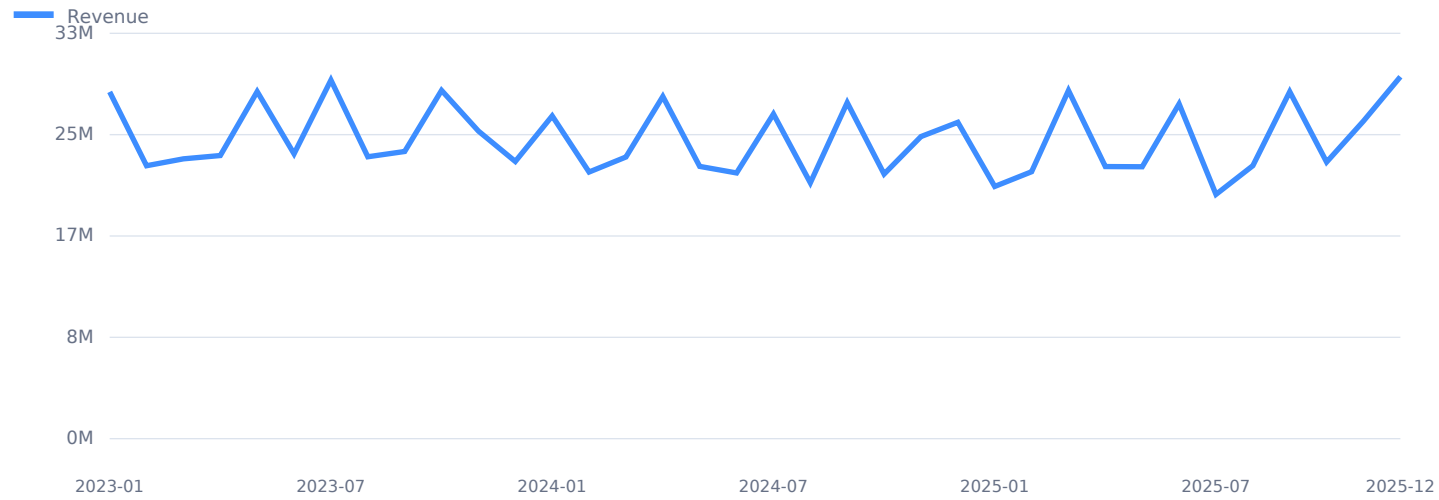
HISTORICAL REVENUE
TRY 891.7M
2023-2025 actual

2026 FORECAST REVENUE
TRY 397.8M
Base scenario

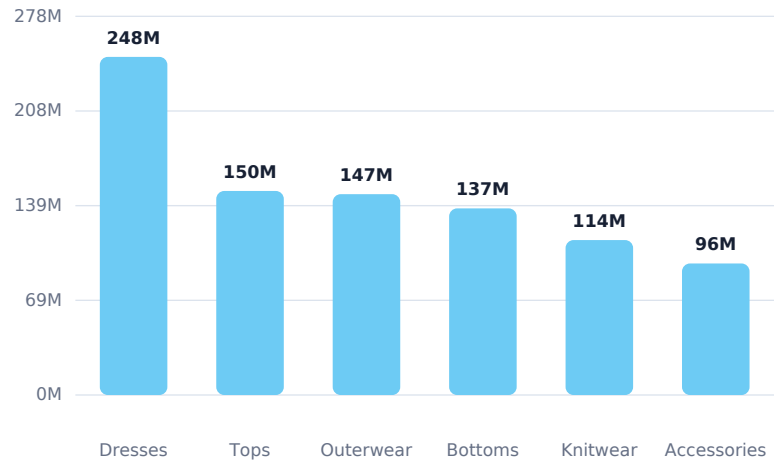
CHAMPION WAPE
15.7%
60 SKU backtest

INVENTORY INVESTMENT
TRY 5.3M
41 SKUs, 10 immediate

Monthly Net Revenue | 2023-2025



Historical Revenue by Category



Executive decision

The portfolio generated TRY 891.7M in synthetic historical revenue. The base plan forecasts TRY 397.8M for 2026 and requires TRY 5.3M of replenishment across 41 SKUs. TRY 2.7M of it is for the 10 SKUs already at or below their reorder point; the rest tops other SKUs up to reorder point plus cycle stock. The champion model portfolio averages 15.7% WAPE.

Demand History & Fulfillment

Where demand was captured and where it was lost

ESTIMATED DEMAND

627,849 units

Uncensored demand proxy

DEMAND FULFILLMENT

86.9%

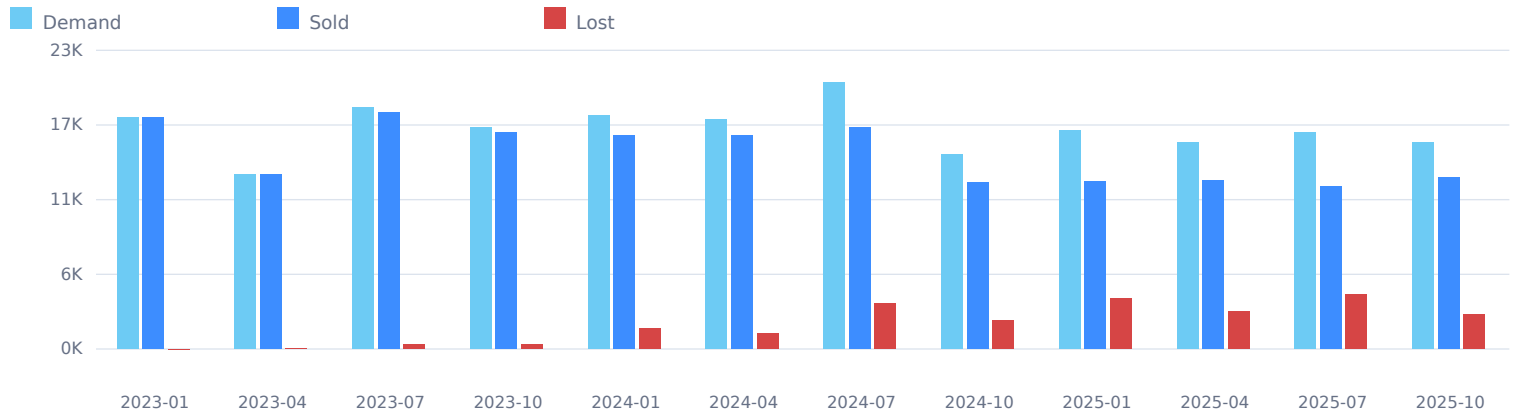
545,753 units sold

LOST SALES

82,096 units

13.1% of demand

Demand, Units Sold, and Lost Sales



Category Fulfillment

Category	Demand	Sold	Fill rate
Dresses	116,273	90,738	78.0%
Bottoms	100,991	83,865	83.0%
Accessories	123,273	108,782	88.2%
Knitwear	98,230	87,046	88.6%
Outerwear	53,817	48,525	90.2%
Tops	135,265	126,797	93.7%

Commercial opportunity

Estimated lost demand equals 82,096 units. Outerwear and seasonal categories exhibit the largest fulfillment pressure, making lead-time planning and safety-stock calibration the highest-impact levers.

Interpretation

Demand is intentionally modeled as units sold plus simulated lost sales. This reduces stockout censoring and gives the forecasting engine a stronger estimate of true customer demand rather than only fulfilled orders.

2026 Demand Forecast

Weekly SKU forecasts aggregated for executive planning

FORECAST UNITS

242,443

52 weeks | 60 SKUs

FORECAST REVENUE

TRY 397.8M

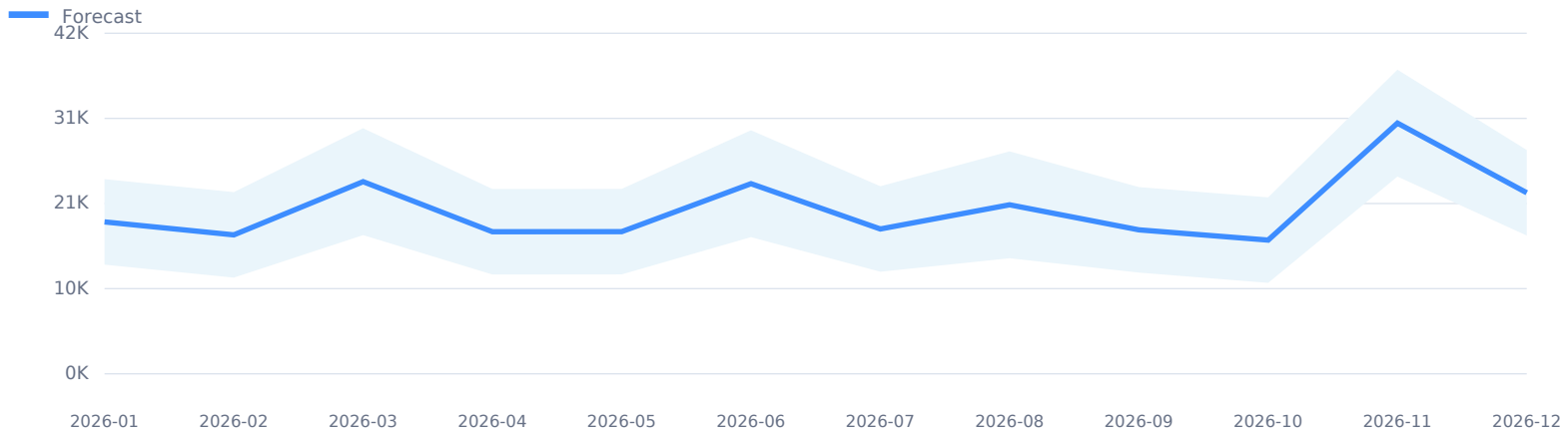
Expected selling price

80% INTERVAL SPAN

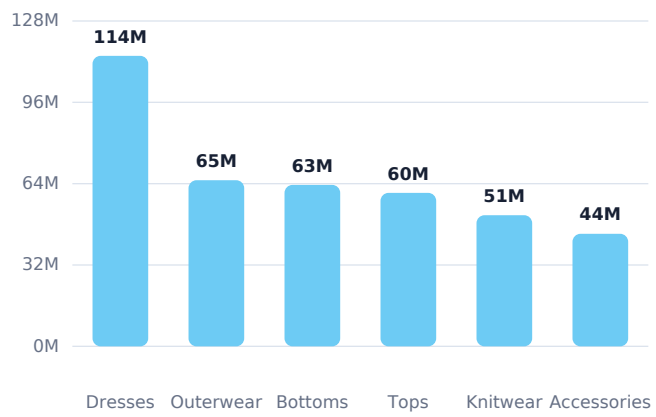
135,954 units

Portfolio uncertainty

Monthly Forecast with 80% Prediction Interval



2026 Forecast Revenue by Category



Planning signal

The 2026 baseline is 242,443 units and TRY 397.8M revenue. Forecast uncertainty is explicitly carried into the safety-stock calculation through SKU-level residual standard deviations.

How to use

Use the point forecast for purchasing and capacity planning; use the 80% interval to stress-test inventory investment, supplier capacity, and campaign exposure before committing budget.

Forecast Model Performance

Backtest governance and champion model selection

AVERAGE CHAMPION WAPE

15.7%

Lower is better

GRADIENT BOOSTING WINNERS

55 SKUs

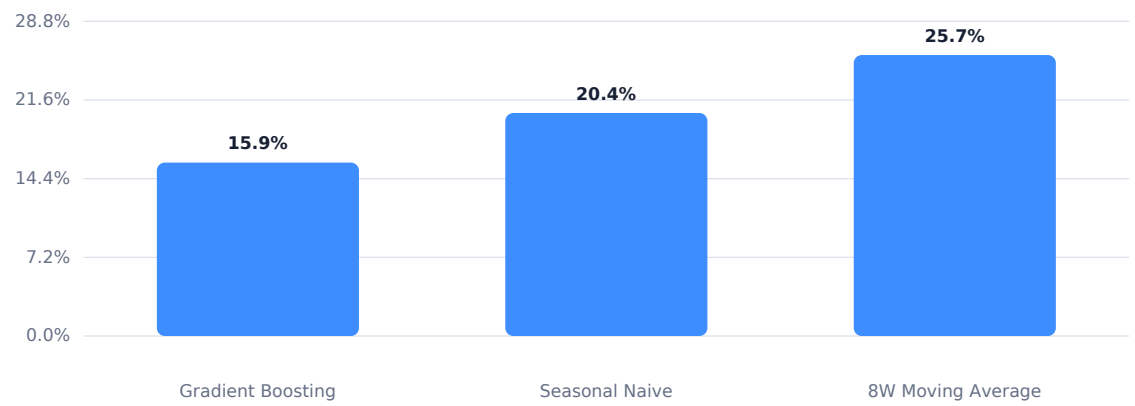
Portfolio majority

BACKTEST HORIZON

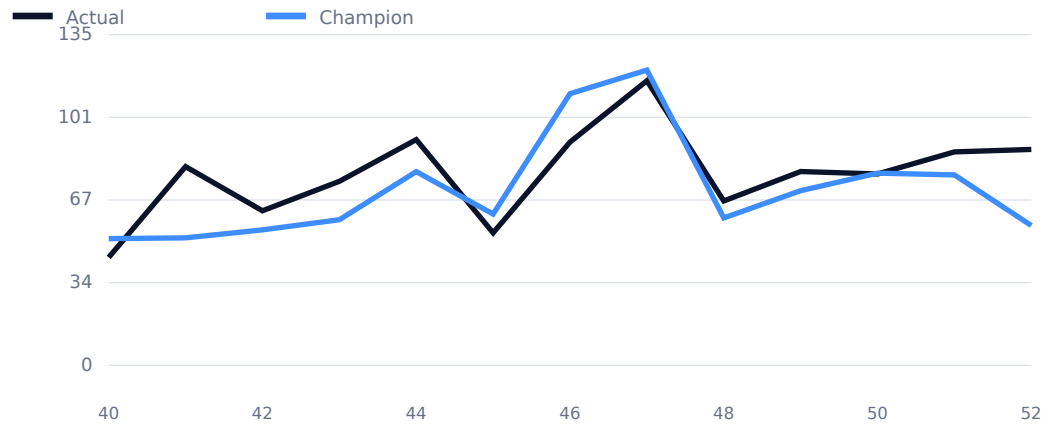
13 weeks

MAE, RMSE, WAPE, Bias

Model Comparison | Average WAPE



Representative SKU Backtest | OUT-002



Model	Avg WAPE	Avg MAE
Gradient Boosting	15.9%	13.5
Seasonal Naive	20.4%	17.3
8-Week Moving Average	25.7%	22.1

Governance rule

Each SKU selects the model with the lowest WAPE on the most recent 13-week holdout. The champion label, residual variability, and bias are persisted so downstream inventory decisions remain auditable.

Inventory Health

Current coverage, action status, and stock exposure

CURRENT INVENTORY VALUE

TRY 11.5M

Latest weekly snapshot

AVERAGE WEEKS OF SUPPLY

5.7 weeks

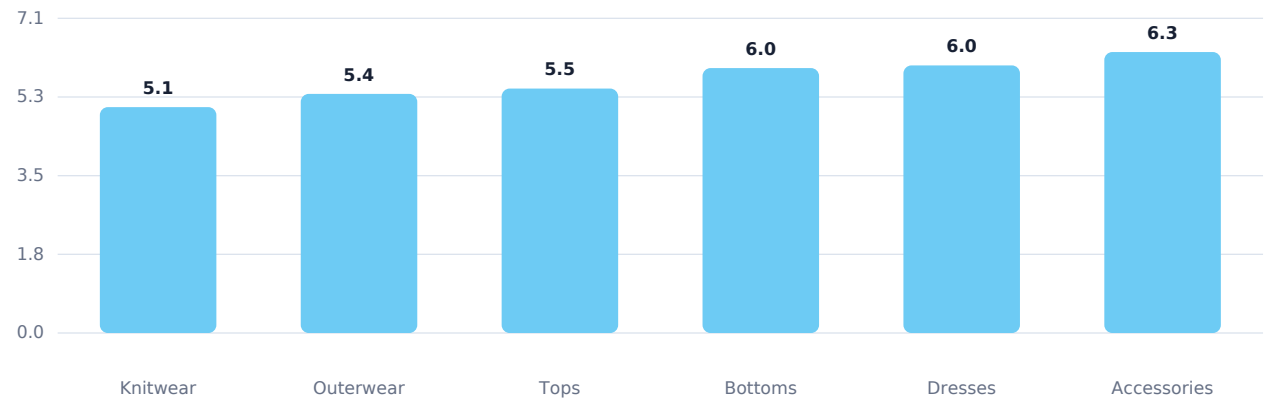
Base forecast

LATEST STOCKOUT SKUS

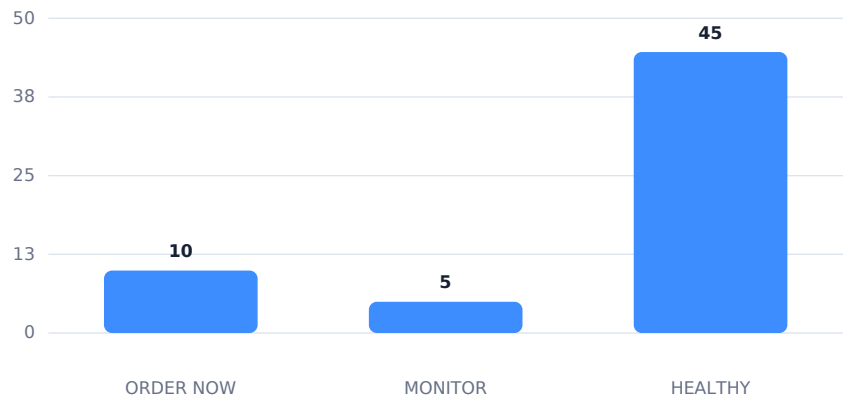
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Simulated snapshot

Projected Weeks of Supply by Category



SKU Action Distribution



SKU	Product	Category	WOS	Action
OUT-009	Urban Outerwear	Outerwear	1.0	ORDER NOW
TOP-003	Linen Top	Tops	1.3	ORDER NOW
DRE-008	Minimal Dresse	Dresses	1.7	ORDER NOW
BOT-006	Cargo Bottom	Bottoms	1.9	MONITOR
DRE-003	Wrap Dresse	Dresses	2.3	ORDER NOW
OUT-005	Cropped Outerwear	Outerwear	2.9	HEALTHY

Replenishment Plan

Pack-rounded purchase recommendations under the base scenario

RECOMMENDED INVESTMENT

TRY 5.3M

41 SKUs with an order

RECOMMENDED ORDER UNITS

7,780

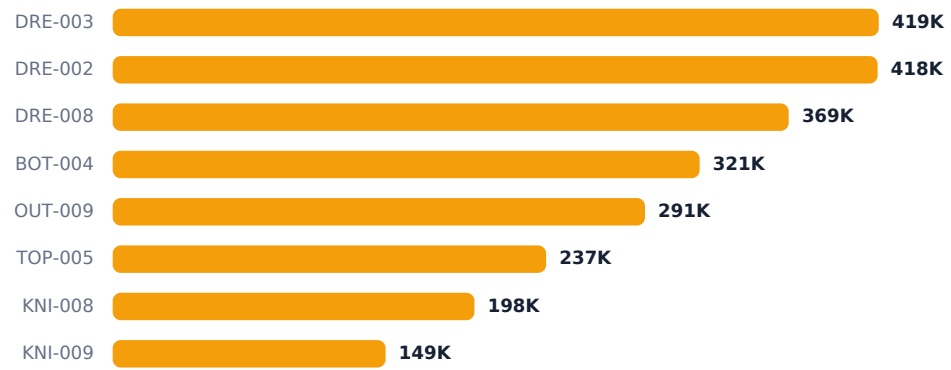
MOQ and case-pack compliant

IMMEDIATE SUPPLIERS

6

Required for ORDER NOW SKUs

Top Immediate Order Investments



Immediate Purchase Orders

SKU	Category	Qty	TRY M	WOS
DRE-003	Dresses	424	TRY 0.4	2.3
DRE-002	Dresses	336	TRY 0.4	5.4
DRE-008	Dresses	370	TRY 0.4	1.7
BOT-004	Bottoms	336	TRY 0.3	3.8
OUT-009	Outerwear	232	TRY 0.3	1.0
TOP-005	Tops	444	TRY 0.2	2.9
KNI-008	Knitwear	444	TRY 0.2	4.6
KNI-009	Knitwear	336	TRY 0.1	3.3

Ordering logic

Recommended quantity is the greater of cycle-stock need, EOQ, and MOQ, then rounded to the product case pack. Inventory position includes both current on-hand stock and open purchase orders.

Operational control

Review ORDER NOW items first, validate supplier capacity, confirm promotion timing, then release purchase orders. MONITOR items should be reassessed weekly as demand and inbound dates update.

Category & SKU Portfolio

Revenue, margin, fulfillment, and concentration

TOP 5 SKU REVENUE SHARE

19.7%

Concentration indicator

PORTFOLIO GROSS MARGIN

58.4%

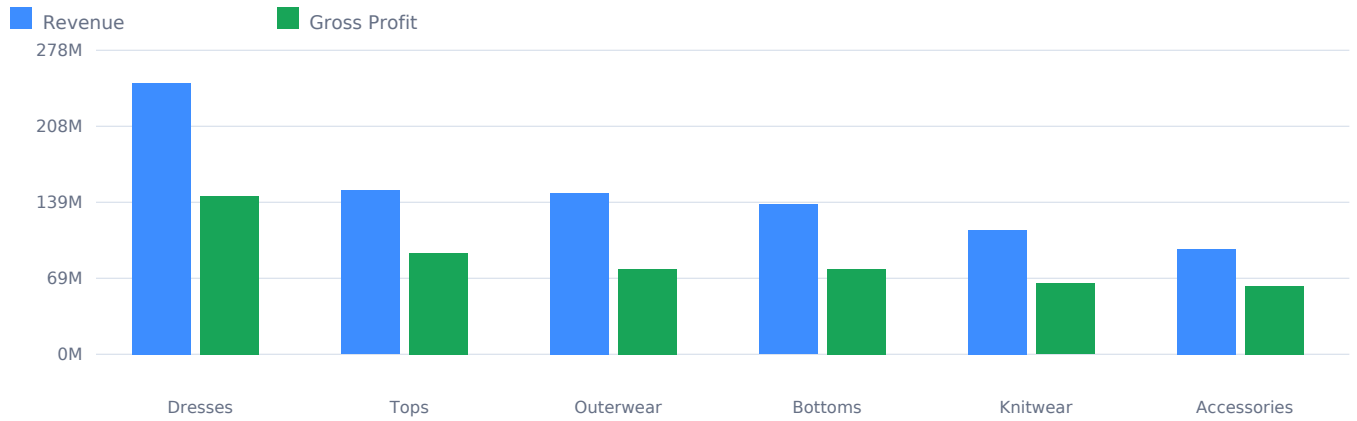
Historical weighted margin

LEADING CATEGORY

Dresses

TRY 247.9M

Revenue and Gross Profit by Category



Top Revenue SKUs

SKU	Product	Revenue	Margin
DRE-006	Ribbed Dresse	TRY 42.0M	55.3%
DRE-004	Linen Dresse	TRY 41.0M	53.9%
DRE-010	Signature Dresse	TRY 34.4M	64.4%
DRE-002	Midi Dresse	TRY 29.6M	58.5%
OUT-007	Quilted Outerwear	TRY 28.7M	52.6%
TOP-006	Oxford Top	TRY 25.0M	61.9%
DRE-008	Minimal Dresse	TRY 24.4M	63.6%
DRE-007	Floral Dresse	TRY 23.6M	58.2%

Portfolio implication

The top five SKUs contribute 19.7% of historical revenue, indicating a diversified portfolio rather than a single-product dependency. Category margin and fulfillment should be reviewed together: high revenue with low fulfillment signals lost-sales opportunity; high stock with weak margin signals working-capital risk.

Supplier Performance

Lead-time reliability and purchasing exposure

PURCHASE VALUE

TRY 373.7M

Historical simulated POs

WEIGHTED ON-TIME RATE

84.2%

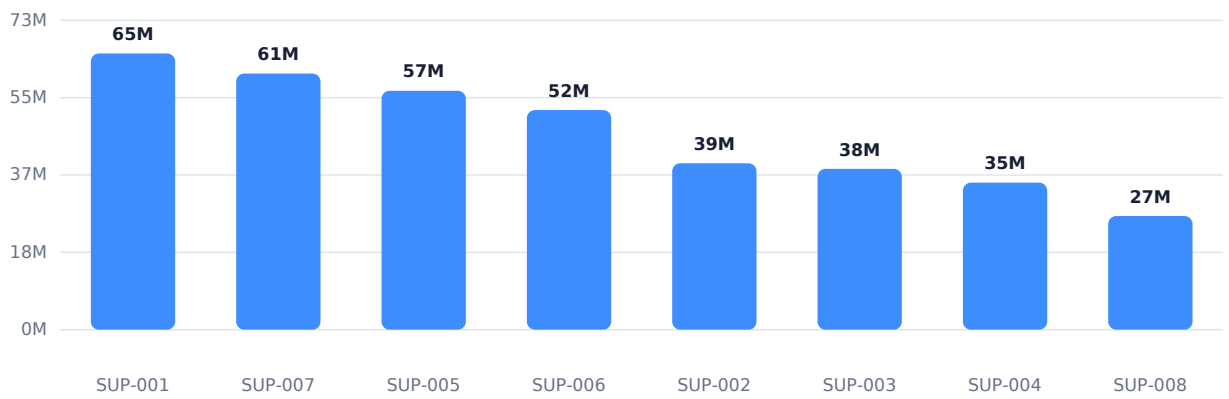
Delivery reliability

AVERAGE ACTUAL LEAD TIME

19.3 days

PO-weighted

Purchase Value by Supplier



Supplier Reliability Matrix

Supplier	POs	Lead	On-time	Value
SUP-001	158	13.7	8.2%	TRY 65.4M
SUP-007	175	27.6	92.6%	TRY 60.6M
SUP-005	176	33.9	96.0%	TRY 56.5M
SUP-006	175	20.4	96.6%	TRY 52.0M
SUP-002	157	20.6	96.2%	TRY 39.4M
SUP-003	155	13.5	98.7%	TRY 38.1M
SUP-004	125	7.0	96.0%	TRY 34.8M
SUP-008	87	7.1	92.0%	TRY 26.9M

Supplier risk

Service-level protection depends on lead-time reliability. The Resilience scenario adds 14 days to supplier lead time, translating delivery uncertainty into higher safety stock and a larger capital requirement.

Procurement action

Use on-time delivery, purchase exposure, and SKU criticality together when allocating supplier capacity. High-exposure suppliers with weaker reliability deserve earlier order release or secondary-source qualification.

Scenario Analysis

Trade-offs across Lean, Base, Growth, and Resilience plans

BASE INVESTMENT

TRY 5.3M

TRY 2.7M immediate (10 SKUs)

GROWTH INVESTMENT UPLIFT

53.5%

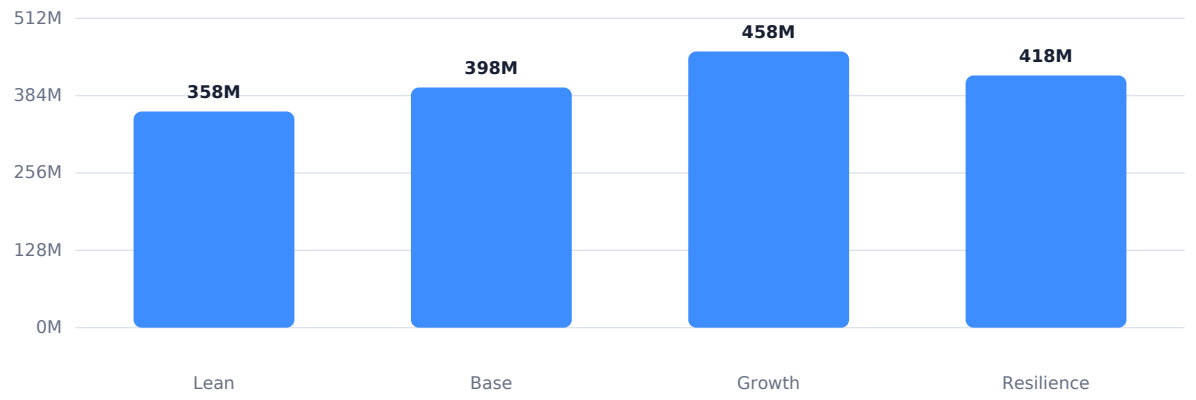
TRY 8.2M

RESILIENCE PREMIUM

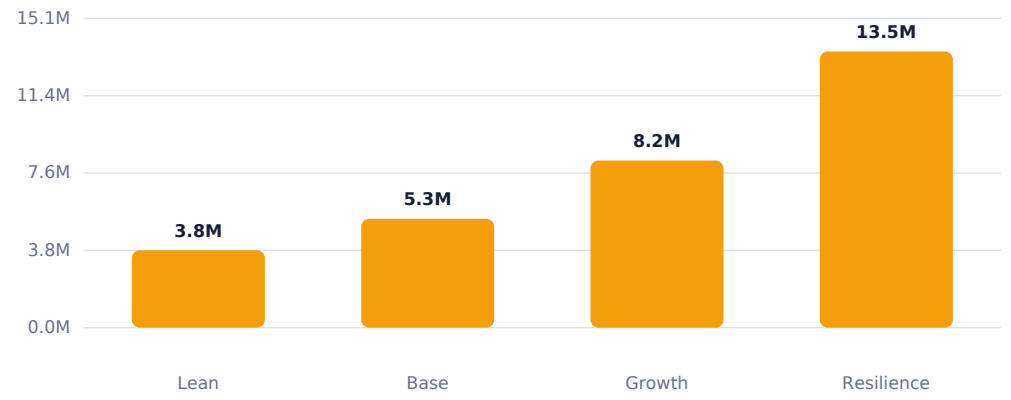
TRY 8.2M

14-day lead-time shock

Forecast Revenue by Scenario



Inventory Investment by Scenario



Scenario	Forecast Units	Investment	SKUs to Order	Avg WOS
Lean	218,199	TRY 3.8M	7	6.3
Base	242,443	TRY 5.3M	10	5.7
Growth	278,809	TRY 8.2M	13	5.0
Resilience	254,565	TRY 13.5M	30	5.4

Decision recommendation

Use Base as the operating plan, Growth as the upside budget request, and Resilience only when supplier-disruption signals emerge.

Executive Recommendations & Roadmap

Turning the analytical model into a repeatable planning process

0-30 DAYS

Release immediate POs
Validate the 10 ORDER NOW SKUs, supplier capacity, case packs, and promotion dates.

Protect critical stock
Prioritize high-revenue SKUs with low projected weeks of supply.

31-60 DAYS

Operationalize forecast review
Refresh actual demand weekly, rerun the champion model, and compare bias by category.

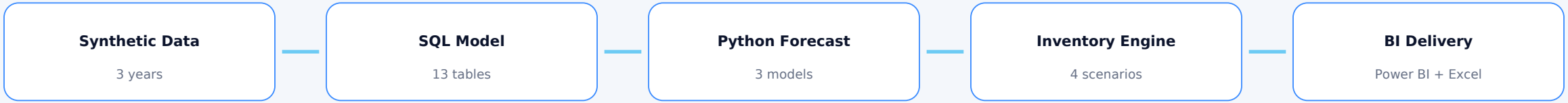
Improve supplier controls
Track on-time rate and actual lead-time variance by supplier.

61-90 DAYS

Move to exception-based planning
Automate alerts for reorder-point breaches, high WAPE, and budget overruns.

Govern scenarios
Approve Base, Growth, or Resilience plans through a monthly S&OP; cadence.

END-TO-END ANALYTICS ARCHITECTURE



Portfolio value

This project demonstrates statistical forecasting, machine learning, SQL engineering, inventory mathematics, executive BI design, scenario planning, testing, and professional documentation within one coherent business decision system.